Supplement 1. Details of electronic database search strategies of the systematic review.

Sources: Pubmed, Embase, Cochrane, CINAHL, Web of Science, SPORTDiscus, Scopus,

Search dates: 18 February 2020.

Search strategies for PubMed:

(1) Ankle-related terms: (ankle* OR talo* OR tibiofib* OR tibio-fib* OR "lateral ligament" OR "lateral ligaments") [Title/Abstract]

(2) Injury-related terms: ("ankle injuries" [MESH] OR unstable OR instabilit* OR strain* OR sprain* OR rupture* OR tear*) [Title/Abstract]

(3) Proprioception-related terms: (propriocep* OR sensor* OR sensa* OR sense* OR feedback* OR movement* OR motion* OR kinetic* OR kinematic* OR kinesthe* OR position*) [Title/Abstract]

(4) Search strategies： (1) AND (2) AND (3)

Supplement 2. Characteristics of studies included in the systematic review.

Authors (Year)	Number of CAI Patients	Number of Healthy Controls	Comparsion	Proprioception Measurement	Measurement Device	Vision Blocked	Foot Bared	Movement Direction (Target Angle)	Angular Velocity	Reliability	Result Display	Summarized Result	Included in Meta-analysis (Reasons for Exclusion)
Garn, SN. et al (1988)	20	/	BL	TTDPM	Kinesthesiometer	Yes	Yes	Ankle Plantarflexion	0.3°/s	NR	Accuracy	+	No (result display differed)
Lentell, G. et al (1995)	42	/	BL	TTDPM	Platform moved by electric motor	Yes	Yes	Ankle Inversion	0.3°/s	NR	Mean value	+	Yes
Konradsen, L. et al (2000)	23	20	BL BG	Active JPR	Torsion goniometer with the foot plate rotated	Yes	Yes	Ankle Inversion (10° 15° 20°)	/	ICC 0.92	AE	+	Yes
Refshauge KM. et al (2003)	39	30	BG	TTDPM	Foot plate coupled to a linear servomotor	NR	Yes	Ankle Inversion Ankle Eversion	0.1°/s 0.5°/s 2.5°/s	NR	70% detection level	+	Yes
Brown, C. et al (2004)	10	10	BG	Passive JPR	Biodex dynamometer	Yes	NR	Ankle Inversion (10%~90% of max) Ankle Eversion (10%~90% of max) Ankle Plantarflexion (10%~90% of max) Ankle Dorsiflexion (10%~90% of max)	2°/s	NR	AE	=	Yes
You, S. H. et al (2004)	4	6	BG	Active JPR	Angle-measuring device	Yes	Yes	Ankle Inversion (15°) Ankle Eversion (10°) Ankle Plantarflexion (25°) Ankle Dorsiflexion (20°)	/	ICC 0.88~0.99	AE	=	Yes
Sekir, U. et al (2007)	24	24	BL	Passive JPR	Isokinetic dynamometer	Yes	With Shoes	Ankle Inversion (10°)	1°/s	ICC 0.86~0.89	AE	+	Yes
Yokoyama, S. et al (2008)	17	17	BG	Passive JPR	3D ankle position analysis system (3D-APAS)	Yes	Yes	Ankle Plantarflexion (-10°~30°) Ankle Inversion (0°~20°)	2-3°/s	ICC 0.62~0.92	CE	+	No (indirect measure through movement analysis)
Tsiganos, G. et al	14	14	BG BL	Active JPR	Isokinetic	Yes	NR	**Knee** Flexion	/	NR	VE	+	No

(2008)					dynamometer			(30° 45° 70°)					(result display differed)
Nakasa, T. et al (2008)	12	17	BG BL	Active JPR	Goniometer footplate	Yes	Yes	Ankle Inversion (5°~30°)	/	NR	AE	+	Yes
Santos, M. J. et al (2008)	21	16	BG BL	Passive JPR	Biodex isokinetic dynamometer	Yes	Yes	Ankle Inversion (30°)	5°/s	NR	AE	=	Yes
Lim, E. C. et al (2009)	25	/	BL	Active JPR TTDPM	Biopressure feedback device	Yes	Yes	Ankle Inversion (5%~75% of max) Ankle Eversion (5%~75% of max)	1°/s 45°/s	ICC 0.83~0.95	CE 70% detection level	=	No (result display differed)
Sefton, J. M. et al (2009)	22	21	BG	Active JPR	Biodex System 3 exercise dynamometer	Yes	NR	Ankle Plantarflexion (30°) Ankle Inversion (15°)	/	NR	CE AE VE	=	Yes
Sefton, J. M. et al (2011)	12	9	BG	Active JPR	Biodex System 3 exercise dynamometer	Yes	Yes	Ankle Plantarflexion (30°) Ankle Inversion (15°)	/	NR	VE	=	No (result display differed)
Witchalls, J. B. et al (2012)	13	8	BG	AMEDA	AMEDA device	NR	With Shoes	Ankle Inversion (10.5°~14.5°)	/	ICC 0.83	AUC	+	Yes
Witchalls, J. B. et al (2014)	25	36	BG	AMEDA	AMEDA device	NR	Yes	Ankle Inversion (10.5°~14.5 °)	/	ICC 0.8	AUC	=	Yes
Iwao, K. et al (2014)	10	20	BG	Active JPR	Goniometer footplate	Yes	Yes	Ankle Inversion (5°~30°)	/	NR	AE	+	Yes
Kim, C. Y. et al (2014)	35	34	BG	Active JPR	Flexible twin axis electrogoniometer	Yes	Yes	Ankle Plantarflexion (0% 20% 30% of max) Ankle Dorsiflexion (0% 20% 30% of max) Ankle Inversion (0% 20% 30% of max) Ankle Eversion (0% 20% 30% of max)	/	ICC 0.75~0.95	AE	+	Yes
Springer, S. et al (2015)	14	14	BG	Active JPR	Biodex Multi-Joint System	Yes	/	**Shoulder** External rotation (45° 60°)	/	NR	AE	=	No (the only study of shoulder)
Lin, C. H. et al (2016)	13	11	BG	Active JPR	Custom-made device: an ankle haptic interface composed of aluminum, two rotary potentiometers,	Yes	NR	Ankle Inversion (10°) + dorsiflexion (10°); Ankle Eversion (10°) + dorsiflexion (10°); Ankle Inversion (10°) + plantarflexion (10°); Ankle Eversion (10°) +	/	ICC 0.99	AE through movement analysis	+	No (indirect measure from movement analysis)

					and a lower extremity support			plantarflexion (10°).					
Hanci, E. et al (2016)	13	/	BL	Active JPR TTDPM	Isokinetic dynamometer	Yes	NR	Ankle Inversion (10° 20°) Ankle Plantarflexion (15° 30°)	0.1°/s	NR	AE Mean value	+	Yes
Kim, C. Y. et al (2016)	40	39	BG	Active JPR	Flexible twin axis electrogoniometer	Yes	YES	Ankle Plantarflexion (10% 20% 30% of max) Ankle Dorsiflexion (10% 20% 30% of max) Ankle Inversion (10% 20% 30% of max) Ankle Eversion (10% 20% 30% of max)	/	NR	AE	+	Yes
Marinho, H. V. R. et al (2017)	18	18	BG	Passive JPR TTDPM	Isokinetic dynamometer i	Yes	Yes	Ankle Plantarflexion (10°)	2°/s	NR	AE Mean value	TTDPM + Passive JPR =	Yes
Saavedra-Miranda, M. et al (2017)	20	/	BL	Active JPR Passive JPR	REV isokinetic instrument	Yes	NR	Ankle Plantarflexion (30°)	5°/s	NR	AE	Passive JPR +	Yes
Sousa, A. S. P. et al (2017)	24	20	BG BL	Active JPR Passive JPR TTDPM	Isokinetic dynamometer	Yes	Yes	Ankle Inversion (5° 15°)	0.25°/s 1°/s	NR	AE Mean value	TTDPM +	Yes
Navarro-Santana, M. J. et al (2019)	28	27	BG	Active JPR	Digital inclinometer	Yes	NR	**Knee** Flexion (45°)	/	NR	AE	=	No (the only study of knee with AE)
Cho, B. K. et al (2019a)	25	/	BL	Passive JPR	Biodex-II isokinetic dynamometer	Yes	Yes	Ankle Inversion (10° 20°)	1°/s	NR	AE	=	Yes
Cho, B. K. et al (2019b)	35	/	BL	Passive JPR	Biodex-II isokinetic dynamometer	Yes	Yes	Ankle Inversion (10° 20°)	1°/s	NR	AE	+	Yes
Nanbancha, A. et al (2019)	19	19	BG	Passive JPR	Biodex isokinetic dynamometer	Yes	Yes	Ankle Inversion (10°) Ankle Eversion (30°) Ankle Dorsiflexion (15°) Ankle Plantarflexion (30°)	5°/s	NR	AE	=	Yes
Lee, J. H. et al (2020)	86	/	BL	Active JPR	Biodex Stability System	Yes	Yes	Ankle Inversion (15°) Ankle Eversion (15°) Ankle Dorsiflexion (15°)	/	NR	Mean angle	=	No (result display differed)

								Ankle Plantarflexion (15°)					

+: Evidence supporting that ankle proprioception deficits existed in CAI patients.
=: Evidence supporting that ankle proprioception did not differ from controls in CAI patients.

CAI, chronic ankle instability; CAIT, Cumberland ankle instability tool; BG, between groups; BL, between limbs; TTDPM, threshold for detection of passive movement; JPR, joint position reproduction; AMEDA, active movement extent discrimination apparatus; NR, not reported; ICC, intraclass correlation coefficients; AE, absolute error; CE, constant error; VE, variable error; AUC, Area under curve.

Supplement 3.1. Quality assessment rating of studies included in the systematic review.

Authors (Year)	Garrn, SN. et al (1988)	Lentell, G. et al (1995)	Konradsen, L. et al (2000)	Refshauge KM. et al (2003)	Brown, C. et al (2004)	You, S. H. et al (2004)	Sekir, U. et al (2007)	Yokoyama, S. et al (2008)	Tsiganos, G. et al (2008)	Nakasa, T. et al (2008)
Hypothesis/aim/objective clearly described.										
Exposure variables clearly described.										
Main outcomes clearly described.										
Study design clearly described.										
Source of subject population clearly described.										
Eligibility criteria for subject clearly described.										
Participation rate reported, ascertainment of record availability described										
Characteristics of study participants described.										
Characteristics of subjects lost after entry/subjects not participating.	?	?	?	?	?	?	?	?	?	?
Important covariates and confounders described.										
Statistical methods clearly described.										
Main findings of the study clearly described.										
Study provide estimates of random variability in data for main outcomes										
Study provide estimates of the statistical parameters.										
Sample size calculations performed and reported.										
Comparison/reference group comparable to the exposed group.										
Participation rate and ascertainment of record availability adequate.										
Study subjects from different groups recruited over same period of time.										
Subject losses or unavailable records were taken account.	?	?	?	?	?	?	?	?	?	?
Diagnosis of ankle instability reliable.										
Diagnosis of ankle instability valid.										
Methods of assessing the exposure variables similar for each group.										
Observers blinded to subject groupings or disease status.										
Main outcome measures reliable.										
Main outcome measures valid.										
Methods of assessing the outcome variables standard across all groups.										
Observation taken over the same time for all groups.										
Prior history of disease and/or injury collected and included in the	?	?	?	?	?	?	?	?	?	?
Adequate adjustment for covariates and confounders.										
Outcome data reported by levels of exposure.										
Outcome/exposure data reported by subgroups of subjects.										
Study results be applied to the eligible population.	?	?	?	?	?	?	?	?	?	?
Study results be applied to other relevant populations.	?	?	?	?	?	?	?	?	?	?
Overall EAI score (0~1)	0.48	0.48	0.48	0.48	0.45	0.52	0.53	0.50	0.50	0.45

Santos, M. J. et al (2008)									?									?								?				?	?	0.48
Lim, E. C. et al (2009)									?									?								?				?	?	0.55
Sefton, J. M. et al (2009)									?									?								?				?	?	0.48
Sefton, J. M. et al (2011)									?									?								?				?	?	0.56
Witchalls, J. B. et al (2012)									?									?								?				?	?	0.47
Witchalls, J. B. et al (2014)									?									?								?				?	?	0.47
Iwao, K. et al (2014)									?									?								?				?	?	0.45
Kim, C. Y. et al (2014)									?									?								?				?	?	0.52
Springer, S. et al (2015)									?									?								?				?	?	0.47
Lin, C. H. et al (2016)									?									?								?				?	?	0.52
Hanci, E. et al (2016)									?									?								?				?	?	0.53
Kim, C. Y. et al (2016)									?									?								?				?	?	0.50
Marinho, H. V. R. et al (2017)									?									?								?				?	?	0.45
Saavedra-Miranda, M. et al (2017)									?									?								?				?	?	0.52
Sousa, A. S. P. et al (2017)									?									?								?				?	?	0.55
Navarro-Santana, M. J. et al (2019)									?									?								?				?	?	0.47
Cho, B. K. et al (2019a)									?									?								?				?	?	0.58
Cho, B. K. et al (2019b)									?									?								?				?	?	0.58
Nanbancha, A. et al (2019)									?									?								?				?	?	0.53
Lee, J. H. et al (2020)									?									?								?				?	?	0.64

Black = Reported, Grey = Partially reported, White = Not reported, “?” = Unable to determine.

Supplement 3.2. Risk of bias assessment of studies included in the systematic review.

Authors (Year)	External validity	Internal validity					
	Performance	Detection	Attrition	Selection bias/control of confounding			
	1	2	3	4	5	6	7
Garn, SN. et al (1988)	✔	✘	✘	✔	✔	✔	✔
Lentell, G. et al (1995)	✔	✔	✘	✔	✔	✔	✔
Konradsen, L. et al (2000)	✔	✘	✘	✘	✔	✔	✔
Refshauge KM. et al (2003)	✔	✘	✘	✘	✔	✔	✔
Brown, C. et al (2004)	✔	✘	✘	✘	✔	✔	✔
You, S. H. et al (2004)	✔	✔	✘	✔	✔	✔	✔
Sekir, U. et al (2007)	✔	✘	✘	✘	✔	✔	✔
Yokoyama, S. et al (2008)	✔	✔	✘	✘	✘	✔	✔
Tsiganos, G. et al (2008)	✔	✔	✘	✘	✔	✔	✔
Nakasa, T. et al	✔	✘	✘	✔	✔	✔	✔

(2008)							
Santos, M. J. et al (2008)	✔	✕	✕	✔	✔	✔	✔
Lim, E. C. et al (2009)	✔	✔	✕	✔	✔	✔	✔
Sefton, J. M. et al (2009)	✔	✕	✕	✕	✔	✔	✔
Sefton, J. M. et al (2011)	✔	✕	✕	✕	✔	✔	✔
Witchalls, J. B. et al (2012)	✔	✕	✕	✔	✔	✔	✔
Witchalls, J. B. et al (2014)	✔	✕	✕	✔	✔	✔	✔
Iwao, K. et al (2014)	✔	✕	✕	✔	✔	✔	✔
Kim, C. Y. et al (2014)	✔	✔	✕	✔	✔	✔	✔
Springer, S. et al (2015)	✔	✕	✕	✕	✔	✔	✔
Lin, C. H. et al (2016)	✔	✕	✕	✕	✕	✔	✔
Hanci, E. et al (2016)	✔	✕	✕	✔	✔	✔	✔
Kim, C. Y. et al (2016)	✔	✔	✕	✔	✔	✔	✔
Marinho, H. V. R. et al	✔	✕	✕	✔	✔	✔	✔

(2017)							
Saavedra-Miranda, M. et al (2017)	✔	✕	✕	✕	✔	✔	✔
Sousa, A. S. P. et al (2017)	✔	✕	✕	✔	✔	✔	✔
Navarro-Santana, M. J. et al (2019)	✔	✕	✕	✔	✔	✔	✔
Cho, B. K. et al (2019a)	✔	✕	✕	✔	✔	✔	✔
Cho, B. K. et al (2019b)	✔	✕	✔	✔	✔	✔	✔
Nanbancha, A. et al (2019)	✔	✔	✕	✔	✔	✔	✔
Lee, J. H. et al (2020)	✔	✕	✕	✕	✔	✔	✔

✔ = the measure was addressed adequately

✕ = the measure was not address adequately or clearly.

1. Performance (Representativeness): ✔ if the study described demographic details (e.g. age and gender), and were representative of "chronic ankle instability" population.
2. Detection (Blinded assessor): ✔ if data assessed or processed by a blinded assessor.
3. Attrition (Participation rate): ✔ if more than 80% of those who volunteered for the study were used eventually.
4. Appropriate description of characteristics of study participants: ✔ if they reported the character of chronic ankle instability (i.e. unilateral/bilateral or both, times of ankle sprain, or duration of symptoms) and control (i.e. no history of ankle sprain).
5. Appropriate ankle proprioception testing protocol: ✔ if they used valid and reliable ankle proprioception testing device (e.g. Isokinetic dynamometer)
6. Appropriate normalization: ✔ if between group comparisons normalized to a standard reference is described (e.g. healthy controls or uninjured contralateral limb)

7. Appropriate statistical tests used: ✔ if describes type of tests (e.g. parametric/nonparametric according to normality, or adjusting for confounding).

Supplement 3.3. CAI patients Inclusion criteria of studies included in the systematic review based on the recommendation of International Ankle Consortium.

Authors (Year)	Population defined by authors	All criteria required				At least 1 of 3 criteria required			Recommended	Percentage of required criteria met
		A history of at least 1 significant ankle sprain	Injury resulting in pain, swelling, interruption of physical activity for at least 1 day	The initial ankle sprain occurred at least 12 months prior to the study	No history of ankle sprain in last 3 months	At least 2 episodes of "giving way" in the last 6 months	Ankle sprain recurrence (at least 2 sprains on same ankle)	Self-reported ankle instability confirmed by questionnaires[a] (AII, CAIT, IdFAI)	Self-reported foot and ankle disability confirmed by questionnaires[b] (FAAM, FAOS)	
Garn, SN. et al (1988)	RS	■					■			40%
Lentell, G. et al (1995)	FAI	■	▒		■	▒				40%
Konradsen, L. et al (2000)	FAI	■					■			40%
Refshauge KM. et al (2003)	RS	■			▒		■			40%
Brown, C. et al (2004)	FAI	■			■	▒	■		▒	60%
You, S. H. et al (2004)	CAI	■				▒				20%
Sekir, U. et al (2007)	FAI	■	▒			■	■			40%
Yokoyama, S. et al (2008)	FAI	■	▒			▒	■			40%
Tsiganos, G. et al (2008)	CAI	■			■	▒	■			60%
Nakasa, T. et al (2008)	CAI	■				▒				20%
Santos, M. J. et al (2008)	FAI	■			▒	▒	■			40%
Lim, E. C. et al (2009)	FAI	■						▒	▒	20%
Sefton, J. M. et al (2009)	CAI	■		■			■		▒	60%
Sefton, J. M. et al (2011)	CAI	■		■	■		■		▒	80%
Witchalls, J. B. et al (2012)	FAI	■			■			▒		40%
Witchalls, J. B. et al	CAI	■						▒		20%

(2014)										
Iwao, K. et al (2014)	FAI									40%
Kim, C. Y. et al (2014)	FAI									60%
Springer, S. et al (2015)	CAI									80%
Lin, C. H. et al (2016)	CAI									20%
Hanci, E. et al (2016)	FAI									40%
Kim, C. Y. et al (2016)	FAI									60%
Marinho, H. V. R. et al (2017)	FAI									40%
Saavedra-Miranda, M. et al (2017)	CAI									40%
Sousa, A. S. P. et al (2017)	FAI&MAI									100%
Navarro-Santana, M. J. et al (2019)	CAI									40%
Cho, B. K. et al (2019a)	FAI									40%
Cho, B. K. et al (2019b)	FAI									40%
Nanbancha, A. et al (2019)	CAI									40%
Lee, J. H. et al (2020)	MAI&FAI									40%

Black = Reported, White = Not Reported, Grey = number or time frame not stated or not achieved the standard recommended IAC.

IAC, International Ankle Consortium; CAI, chronic ankle instability; FAI, functional ankle instability; MAI, mechanical ankle instability.

[a] Ankle Instability Instrument (AII), Cumberland Ankle Instability Tool (CAIT), and Identification of Functional Ankle Instability (IdFAI).
[b] Foot and Ankle Ability Measure (FAAM), Foot and Ankle Outcome Score (FAOS).